

DIFFERENTIAL EQUATIONS

Lecture 3 — ICs, BCs, Forming DEs & Direction Fields

Initial and Boundary Conditions

When we solve a DE, the general solution contains n arbitrary constants (where n is the order). To find a **unique particular solution**, we need exactly n additional conditions. These come in two types.

Definition: Initial Conditions (ICs)

Conditions in which **all** specified values of the dependent variable and its derivatives are given at **one single value** of the independent variable.

$$\text{Example (2nd order): } y(0) = 1, \quad y'(0) = 2$$

Both conditions are at $x = 0$ — the same point.

Definition: Boundary Conditions (BCs)

Conditions in which the specified values are given at **more than one value** of the independent variable.

$$\text{Example (2nd order): } y(0) = 1, \quad y(1) = 3$$

Conditions are at two different points $x = 0$ and $x = 1$.

Initial Conditions	Boundary Conditions
All conditions at one value of x	Conditions at more than one value of x
$y(0) = 1, y'(0) = 2, y''(0) = 4$	$y(0) = 1, y(1) = 2$
Leads to an Initial Value Problem (IVP)	Leads to a Boundary Value Problem (BVP)

Definition: IVP and BVP

- **Initial Value Problem (IVP):** A DE together with initial conditions.
- **Boundary Value Problem (BVP):** A DE together with boundary conditions.

Eliminating Arbitrary Constants — Forming a DE

Given a family of curves (general solution) with n arbitrary constants, we can work **backwards** to recover the DE it satisfies. This is done by differentiating n times and eliminating all n constants.

Method: Eliminating Arbitrary Constants

1. Write the given relation $f(x, y, C_1, C_2, \dots, C_n) = 0$.
2. Differentiate with respect to x repeatedly until you have $n + 1$ equations (the original plus n derivatives).
3. Use these equations to eliminate all n arbitrary constants algebraically.
4. The result is the DE satisfied by the family of curves.

The resulting DE will be of order n , matching the number of constants.

Example 1: Find the DE for $y = A \sin x + B \cos x$ [2 constants $\Rightarrow$ 2nd order]

Step 1. Write the relation and differentiate twice:

$$y = A \sin x + B \cos x \quad (\text{I})$$

$$y' = A \cos x - B \sin x \quad (\text{II})$$

$$y'' = -A \sin x - B \cos x \quad (\text{III})$$

Step 2. Notice that equation (III) = $-(\text{equation I})$:

$$y'' = -(A \sin x + B \cos x) = -y$$

Step 3. Both constants A and B are eliminated automatically:

$$\boxed{\frac{d^2 y}{dx^2} + y = 0}$$

Example 2: Find the DE for $y = Ce^{2x}$ [1 constant $\Rightarrow$ 1st order]

Step 1. Differentiate once:

$$y = Ce^{2x} \quad (\text{I})$$

$$\frac{dy}{dx} = 2Ce^{2x} \quad (\text{II})$$

Step 2. From (I): $C = ye^{-2x}$. Substitute into (II):

$$\frac{dy}{dx} = 2y e^{-2x} \cdot e^{2x} = 2y$$

Step 3. The constant C is eliminated:

$$\boxed{\frac{dy}{dx} = 2y}$$

Example 3: Find the DE for $y = C_1 e^x + C_2 e^{-x}$ [2 constants $\Rightarrow$ 2nd order]

Step 1. Differentiate twice:

$$y = C_1 e^x + C_2 e^{-x} \quad (\text{I})$$

$$y' = C_1 e^x - C_2 e^{-x} \quad (\text{II})$$

$$y'' = C_1 e^x + C_2 e^{-x} \quad (\text{III})$$

Step 2. Notice equation (III) = equation (I):

$$y'' = y$$

Step 3. Both constants eliminated:

$$\frac{d^2y}{dx^2} - y = 0$$

Example 4: Find the DE for the family of circles $(x - c)^2 + y^2 = c^2$ [1 constant]

Step 1. Expand:

$$x^2 - 2cx + c^2 + y^2 = c^2 \implies x^2 - 2cx + y^2 = 0 \implies c = \frac{x^2 + y^2}{2x} \quad (\text{I})$$

Step 2. Differentiate the original implicitly w.r.t. x :

$$2(x - c) + 2y \frac{dy}{dx} = 0 \implies x - c = -y \frac{dy}{dx} \implies c = x + y \frac{dy}{dx} \quad (\text{II})$$

Step 3. Equate (I) and (II) to eliminate c :

$$\frac{x^2 + y^2}{2x} = x + y \frac{dy}{dx}$$

$$x^2 + y^2 = 2x^2 + 2xy \frac{dy}{dx}$$

$$2xy \frac{dy}{dx} = y^2 - x^2$$

Direction Fields

Definition: Direction Field (Slope Field)

A **direction field** is a graphical representation of a first-order DE $dy/dx = f(x, y)$. At each point (x, y) in the plane, a short line segment is drawn with slope equal to $f(x, y)$. The collection of all these segments gives a visual picture of how solutions behave **without actually solving the DE**.

How to Construct a Direction Field

Constructing a Direction Field for $dy/dx = f(x, y)$

1. Choose a grid of points (x_i, y_j) over the region of interest.
2. At each point, compute the slope $m = f(x_i, y_j)$.
3. Draw a short line segment through (x_i, y_j) with that slope.
4. Solution curves (called **integral curves**) flow along these segments — they are always tangent to the slope marks.

Direction fields are especially useful when:

- The DE cannot be solved analytically.
- You want a quick qualitative picture of all solutions at once.
- You want to understand the long-term behaviour of solutions without computing them.

The slope marks give **local** information (the slope at each point). Following them continuously

gives you the **global** solution curve.

Worked Example

Example: Direction field for $dy/dx = y - x$

At a few sample points, the slope $m = y - x$:

Point (x, y)	Slope $m = y - x$	Direction
(0, 0)	0	horizontal
(0, 1)	1	slope up at 45°
(1, 0)	-1	slope down at 45°
(1, 2)	1	slope up at 45°
(2, 2)	0	horizontal
(0, -1)	-1	slope down

The line $y = x$ (where $m = 0$) forms a **nullcline** — solution curves cross it horizontally. Above it ($y > x$) solutions slope upward; below it ($y < x$) they slope downward.

The general solution of $dy/dx = y - x$ is $y = x + 1 + Ce^x$. Each value of C gives one integral curve that everywhere follows the direction field.

Types of First-Order ODEs — Overview

First-order ODEs are classified by their form, which determines the solution method. The main types covered in this course are:

No.	Type	Characteristic Form
1	Separable ODEs	$dy/dx = F(x)/G(y)$ — variables can be fully separated
2	Reducible to Separable	Homogeneous form $dy/dx = f(y/x)$, solved via $u = y/x$
3	Exact ODEs	$M dx + N dy = 0$ with $\partial M/\partial y = \partial N/\partial x$
4	Reducible to Exact	Non-exact equations made exact via an integrating factor
5	Linear ODEs	$dy/dx + P(x)y = Q(x)$

Each type requires a different technique. The first step when solving any first-order ODE is always to **identify** which type it is — this determines which method to apply. These types are covered in detail in Lectures 4 through 7.

Summary

Concept	Key Point
Initial Conditions	All conditions at one value of the independent variable
Boundary Conditions	Conditions at more than one value of the independent variable
IVP / BVP	DE + ICs = IVP; DE + BCs = BVP
Forming a DE	Differentiate n times and eliminate all n arbitrary constants
Order of resulting DE	Always equals the number of arbitrary constants eliminated
Direction Field	Grid of slope segments from $dy/dx = f(x, y)$; shows solution behaviour without solving
Nullcline	Curve where $f(x, y) = 0$; solution curves cross it horizontally

End of Lecture 3